

Continual Safety Alignment in Vision-Language Models

via Multimodal Gradient-Based Sample Selection

Extending Continual Safety Alignment (Bach et al., 2026) to Multimodal Architectures (<https://arxiv.org/abs/2604.17215v1>)

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The Problem: Alignment Drift in Continual Fine-Tuning

1. Confirmed in Text LLMs (Bach et al., 2026)

- Benign fine-tuning unintentionally erodes safety guardrails as a structural property.
- High-gradient samples drive elastic reversion toward unaligned pretraining distributions.
- Standard remedies like gradient clipping fail (ASR remains ~31% at clip=0.5), because clipping dampens step size without altering alignment-reversing directions.

2. Our Research Question for VLMs

- Does the same mechanism govern Vision-Language Models?
- Bach et al. explicitly leave this open (Limitations): 'Extending gradient-based selection to VLMs requires further investigation.'
- The Multimodal Attack Surface: Unlike text LLMs, VLMs introduce an additional visual attack surface that may not be protected by text-channel alignment where attackers bypass text guardrails via visual features. (Nie et al., 2024)

LLM Elasticity

This project does not assume dataset size alone determines update direction

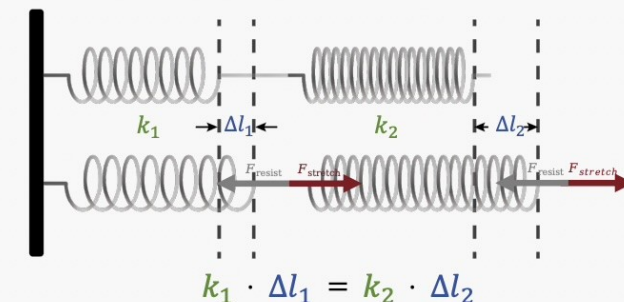
$$F_{\text{elastic}} \propto |\mathcal{D}_i| \cdot \Delta D_{\text{KL}}(\bar{p}_\theta \| \bar{p}_{\mathcal{D}_i})$$

([Ji et al., 2024](#)) shows that LLMs exhibit elasticity: a tendency to revert toward pretrained distributions during fine-tuning because the massive pretraining corpus exerts stronger influence than smaller alignment datasets.

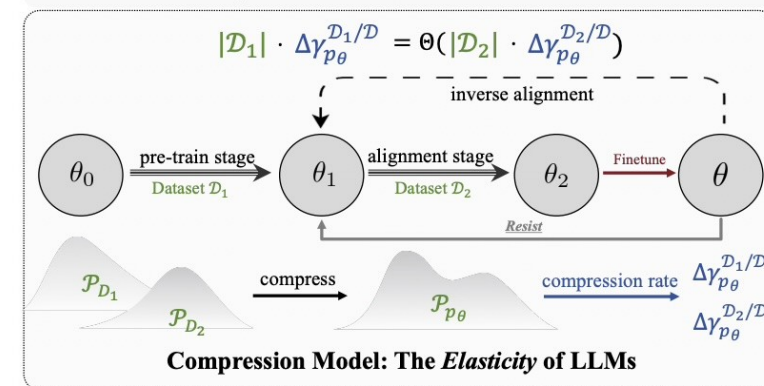
- F_{elastic} is the strength of the dataset's tendency to pull the model toward its own behavior.
- $|\mathcal{D}_i|$ is the size of dataset $\mathcal{D}_i$
- p_θ is the model's current output distribution.
- $p_{\mathcal{D}_i}$ is the behavioral or token distribution represented by dataset $\mathcal{D}_i$
- ΔD_{KL} measures how different the model's current behavior is from that dataset's distribution.
- Since $|\mathcal{D}_p| \gg |\mathcal{D}_a|$ (pre-trained vs aligned data) => **Elastic reversion**

Our Contribution: The *Elasticity* of LLMs

Language models, fine-tuned with perturbations, exhibit an inverse relationship between **normalized compression rate changes** and **dataset volume**, akin to that of a series spring system, revealing the *elasticity* of LLMs.



Physical Model: The Hooke's Law



Compression Model: The *Elasticity* of LLMs

Figure 1: **The *Elasticity* of Language Models.** The change in normalized compression rates ($\Delta \gamma_{p_\theta}^{\mathcal{D}_i/\mathcal{D}}$) and the dataset volume ($|\mathcal{D}_i|$) follows an inverse proportionality law after perturbations, which is akin to the relationship between spring deformation (Δl_i) and stiffness (k_i) in coupled springs. We conjecture that the *elasticity* causes language models to resist alignment, enabling the possibility of *inverse alignment*.

VISAGE: Volumetric Index for Safety Alignment Guided by Explanation

A critical empirical finding is that safety basins have sharp boundaries: safety exhibits step-function collapse when crossing the boundary, with minimal graceful degradation. This geometry makes large parameter updates particularly dangerous.

- **The Safety Basin:** the average depth of a local region in the model's parameter space where the model remains safe.
- $B = \{\theta: S(\theta) \leq S_{threshold}, \theta \text{ connected to } \theta_{align}\}$
- A higher VISAGE score
 - model is safe in its current state
 - is also highly robust
 - less likely to have its safety alignment broken by fine-tuning or adversarial attacks

A. Safety basin universally appears in open-source LLMs' parameter spaces. Randomly perturbing model weights maintains safety level of original aligned model (light purple dot) in its local neighborhood.

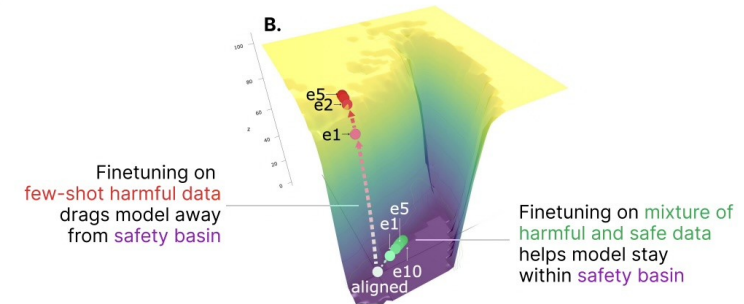
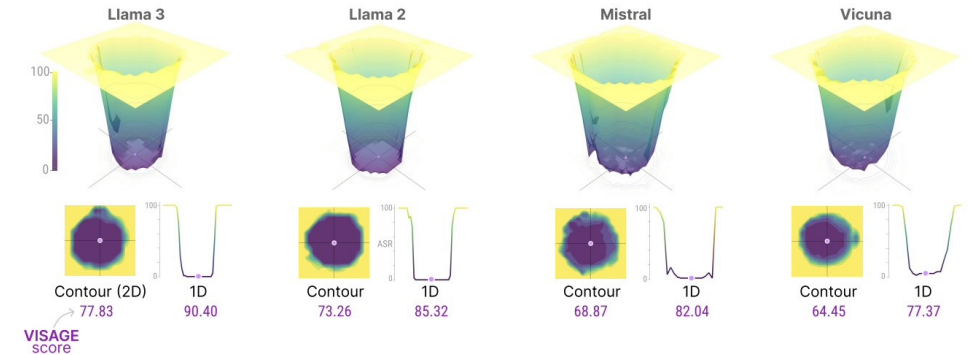


Figure 1: **A.** “Safety basin”, a new phenomenon observed universally in the model parameter space of popular open-source LLMs. Our discovery inspires us to propose the new VISAGE safety metric that measures the safety in LLM finetuning by probing its safety landscape. **B.** Visualizing the safety landscape of the aligned model also enables us to understand why finetuning with harmful data compromises safety but finetuning with both harmful and safe data preserves the safety.

$$\text{VISAGE} = \mathbb{E}_{\alpha \sim U(-a, a)} [S_{\max} - S(\alpha)] \quad \text{s.t. } S < S_{\max}$$

• *Safety / attack success rate (AdvBench, HarmBench)*

VLM Replacement	What it is	LLM benchmark used
MM-SafetyBench	13 safety-critical scenarios with 5,040 text-image pairs. Complementary benchmarks.	AdvBench
FigStep	Embeds harmful instructions as image text alongside a benign text prompt (preserves diverse attack vectors)	No direct text-only equivalent. Attacks via <u>visual OCR channel</u> , tests whether visual text can bypass text-channel safeguards. Distinct attack shape.
JailBreakV-28K	Large-scale multimodal benchmark with 16 harm scenarios across 28k samples– mixing direct, contextual, and optimization-based attacks.	HarmBench
HarmBench (Text Slice)	Text-only attack queries to evaluate vulnerabilities in the VLM's language decoder. 510+ queries, Llama-Guard-3-8B judge	HarmBench

• *Truthfulness /Hallucination (TruthfulQA)*

MMHal-Bench	Primary open-ended, model-graded hallucination benchmark. 8 question types per image across 12 images types	TruthfulQA
POPE	Secondary check using simpler yes/no object-hallucination probing.	TruthfulQA (Supplementary signal)

• *Downstream task sequence (Dolly, GSM8K, MedMCQA, SQuAD v2)*

LLaVA-Instruct-150K	Benign general visual instruction-following to reduce excessive refusal before task-specific tuning.	Dolly
MathVista	Visual mathematical reasoning dataset. 6141 samples from 28 multimodal datasets	GSM8K
VQA-RAD and SLAKE	Medical visual question answering dataset. ~315 images, 3.5k Q/A pairs AND ~642 images, 14k Q/A pairs	MedMCQA
DocVQA	50k Document image reading comprehension and QA, 12k+ document images	SQuAD v2

VLM Attack Surface & Multi-Channel Evaluation

Attack surface	Example
Text channel	HarmBench or AdvBench prompts
Image-text rendering	FigStep-style image instructions
Visual semantic content	MM-SafetyBench
Optimization or jailbreak transfer	JailBreakV-style attacks
Cross-modal conflict	benign text + harmful image, or vice versa
OCR pathway	harmful text rendered in different fonts, layouts, or rotations

You should not rely on one aggregate **ASR (Percentage of attack examples that produce an unsafe response)**
Report:

$$ASR_{text} , ASR_{image} , ASR_{cross} , ASR_{total}$$

This can reveal cases where text safety remains stable while visual safety collapses. Lower is better.

Related Multimodal safety approach

Reference: Nie et al., *Safety Alignment for Vision Language Models*, [arXiv:2405.13581](https://arxiv.org/abs/2405.13581) (SafeVLM)

What SafeVLM Does (The Architectural Approach):

- Adds custom modules:
 - **Safety Projector**,
 - **Safety Tokens**, and
 - **Safety Head**.
- Requires a complex two-stage training process and alters model inference topology.

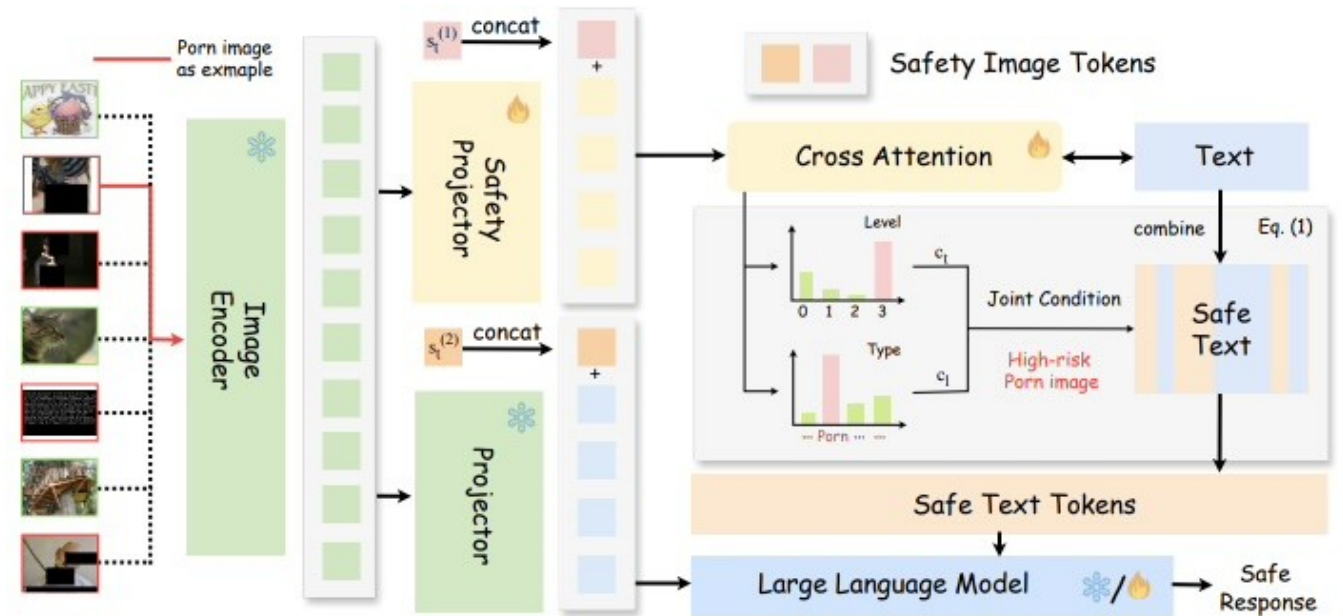


Figure 1: We conduct SafeVLM through two stages of training: (1) freezing the LLMs while learning safety features and adapting these features to align with LLM input, and (2) unfreezing the LLMs to enhance their understanding of unsafe content. During inference, we conditionally mix safety prompts with original inputs.

SafeVLM Empirical Results: Safety Scores & Over-Refusal Trade-offs (Tables 3 & 4).

Their Results on LLaVA-v1.5-7B

1. AdvBench (Vanilla ASR):
 - Reduced from **6.45%** -> **1.72%**.
2. AdvBench (Suffix Injection):
 - Reduced from **78.27%** -> **67.56%**.
3. [XSTest](#) Unsafe Rate:
 - Dropped from **26.50%** -> **7.46%**
(shows significant over-refusal / false refusal risk)
4. [RTVLM](#) Safety Score:
 - Increased from **6.27** -> **8.26**.

Table 3: **Comparison of text attack ASR (%) between the baseline model and the SafeVLM.** The ↓ denotes the ASR is lower the better and the ↑ denotes the ASR is higher the better.

Method	AdvBench		XSTest	
	Vanilla Instruction (↓)	Suffix Injection(↓)	Safe Instruction(↑)	Unsafe Instruction(↓)
LLaVA-v1.5-7B	6.45	78.27	91.20	26.50
SafeVLM	1.72 (↑4.73)	67.56(↑10.71)	76.89(↓14.31)	7.46(↑19.04)
SafeVLM (+LoRA)	1.90 (↑4.55)	69.86(↑8.41)	78.09(↓13.11)	6.96 (↑19.54)

Table 4: **Evaluation on the multimodal benchmarks**, including MMBench (Liu et al., 2023c), SEEDBench(Li et al., 2023b), and MME (Fu et al., 2024).

7B Method	MMBench	SEEDBench	MME ^P	MME	RTVLM
LLaVA-v1.5 7B	64.3	61.6	1487.9	1773.6	6.27
LLaVA-v1.5 7B + RT SFT	66.8	-	-	-	6.88
SafeVLM	66.8	65.3	1479.5	1762.7	8.18
SafeVLM (+LoRA)	68.5	63.7	1458.8	1753.8	8.26

VLM extension: Methodology

- Text-only LLMs, G_i is calculated across LLM parameters or LoRA parameters.
- Modification In VLMs, you have three distinct module groups:
 - Vision Encoder (usually frozen during LoRA)
 - Vision-Language Projector / Connector (trainable cross-attention or MLP)
 - Language Model Backbone (trainable LoRA)
- If you compute gradients across everything, the vision encoder's high visual variance will completely swamp the gradient norm with image noise.

VLM extension: Methodology

- We formalize **Continual Safety Alignment for VLMs** across a sequence of T multimodal tasks $\{D_1, D_2, \dots, D_T\}$
- Where $D_t = \{(x_i^v, x_i^t, y_i)\}_{i=1}^{N_t}$ contains visual inputs, text instructions, and target responses
- For a VLM, replace the language-model parameters with multimodal parameters:

$$\theta^{VLM} = (\theta_{vision_encoder}, \theta_{projector}, \theta_{LM_LoRA})$$

- VLM VISAGE is evaluated in a specified trainable parameter subspace, not automatically in the full model parameter space.
- If training only adaptors:

$$\theta^{train} = (\theta_{LM_LoRA}, \theta_{projector})$$

With a vision encoder and base language model frozen

$$\Rightarrow \theta_t^{VLM} = \arg \min_{\theta} L_t^{VLM}(\theta; D_t^{image, text}) \text{ s.t. } \theta_t^{VLM} \in B_{VLM}$$

- We restrict both sides of the measurement: response-only labels define the loss, and trainable-module selection defines the parameter space in which the gradient norm is measured.

VLM extension: Methodology

Disentangling Modality Tension (Format Mismatch vs. Visual Cross-Attention)

- The original paper identified high- G_i samples in text LLMs as format mismatches (short classification targets vs. verbose aligned outputs).
- In VLMs, high gradients arise from two distinct sources:
 - Output format mismatch (same as LLMs)
 - Visual grounding conflict: Complex images with dense visual information generate high gradient norms because the vision connector projections collide with LLM text priors.
- *Improvement:* Track G_i split by token types if possible, or log sample image complexity (e.g., image resolution, object count) during your Phase 2 audit to see if Moderate- G_i filtering preferentially removes complex visual scenes or standard format mismatches.

VLM extension: Methodology

- A VLM input contains image tokens, system instructions, user prompts, and the target assistant response. During supervised fine-tuning, the loss should be computed **only on the target response tokens**.
- Set labels for all non-target positions to -100 (PyTorch's cross-entropy loss ignores these positions):
 - image-token positions
 - system-message tokens
 - user-prompt tokens
 - special or padding tokens
- Per-sample response loss
$$L_i = -\frac{1}{|Y_i|} \sum_{i \in Y_i} \log p_{\theta}[y_{i,t} | v_i, x_i, y_{i,<t}]$$
 - $v_i = \text{image}$
 - $x_i = \text{text instruction}$
 - $y_i = \text{assistant target}, Y_i(\text{target}) = \text{only assistant response token positions}$
 - Gradients use this Loss

VLM extension: Methodology

Proposed gradient attribution ablation

- This ensures that G_i measures the parameter change associated with learning the target response, rather than learning to reproduce the prompt, special tokens, or padding.
- **Our Adaptation:** We freeze the Vision Encoder and formulate **3 distinct gradient attribution modes**:
 - $G_i^{(L)} = \|\nabla_{\theta_{LM_{LoRA}}} L_i\|_2$
(Language: Evaluates whether alignment drift is driven by conversational response priors in the language model.)
 - $G_i^{(P)} = \|\nabla_{\theta_{projector}} L_i\|_2$
(Projector: Evaluates whether drift is driven by cross-modal feature warping justified by SafeVLM's finding that pre-trained projectors lack safety alignment)
 - $G_i^{(J)} = \sqrt{(G_i^{(L)})^2 + \lambda \cdot (G_i^{(P)})^2}$
(Joint: combines both VLM signals with a balancing scalar)

VLM extension: Methodology

- Scalar (λ) Options include:
 - $\lambda = 1$ after normalized module norms
 - choose λ using a validation set
 - normalize each module norm by its median or standard deviation before combining.
- A safer initial formulation is:
 - $\bar{G}_i^{(J)} = \sqrt{\left(\frac{G_i^{(L)}}{\text{median}(G^{(L)})}\right)^2 + \left(\frac{G_i^{(P)}}{\text{median}(G^{(P)})}\right)^2}$
 - Prevents larger parameter group dominating the joint score because of more parameters

VLM extension: Methodology

VLM-specific diagnostic

◦ Log the following for every sample:

- target-response token count
- prompt-token count
- image-token count
- image resolution
- mean response-token loss
- $G_i^{(L)}$
- $G_i^{(P)}$
- combined gradient score

VLM extension: Methodology

Language LoRA, Projector, Joint Multimodal Tension, Length normalization

- $G_i^{(L)} = \sqrt{G_1^2 + G_2^2 + \dots + G_k^2}$
 - This is the L2 norm (magnitude). It gives one single number showing how hard the language decoder is being pushed.
- $G_i^{(P)} = \sqrt{G_1^2 + G_2^2 + \dots + G_m^2}$
 - Shows how hard the visual-to-language bridge is being pushed by this image.
- $G_i^{(J)} = \sqrt{G_i^{(L)^2} + \lambda \cdot G_i^{(P)^2}}$
 - combining both tensions into one overall VLM score.
- *Normalized* $G_i = \frac{G_i^{(J)}}{\sqrt{\text{target token count}}}$ **(gradient per token)**
 - Prevents long answers from getting high scores just because they have more words.
 - Used for sensitivity analysis alongside mean-response token as primary loss.
 - we will record target length and test length-normalized alternatives

VLM extension: Methodology

- Also record the combined norm:
 - $G_i^{total} = ||\nabla_{\theta_{LM_{LoRA}}, \theta_{projector}} L_i ||_2$
- One total norm will not know which mechanism is responsible => a high gradient could be caused by:
 - a text-output format mismatch
 - an image that is difficult to interpret
 - a poorly grounded visual representation
 - an unusually long target
- Examples groups for first set:
 - Low-G: lowest 20%
 - Moderate-G: samples closest to the median
 - High-G: highest 20%
 - Random: random 20%
 - All: all available samples
- The method is not simply take the lowest gradients. It first *removes extreme-loss* examples, *computes gradient norms* on the remaining candidates, and *selects samples nearest the median gradient norm*.

Execution Plan

Stage	LLM Task	Our VLM Equivalent	Reasoning	Citation
1	Dolly (15K instruction following)	LLaVA-Instruct-150K	General visual instruction following , selected as multimodal analogue to Dolly. Benign, diverse, general purpose. We will measure whether it changes refusal rate, task capability, and safety before using it as the first continual-learning stage.	Liu et al., arXiv:2304.08485
2	GSM8K (math word problems)	MathVista (6,141 problems)	Visual-mathematical reasoning from charts, diagrams, geometry. Direct multimodal analog to GSM8K. Use a documented training subset for adaptation and a disjoint held-out split for evaluation.	Lu et al., arXiv:2310.02255
3	MedMCQA (medical MCQ)	VQA-RAD (external evaluation to test transfer) and SLAKE (preferred for training)	Clinical image understanding: radiology, pathology. VQA-RAD = radiology QA pairs. SLAKE = bilingual medical VQA	Lau et al., Nature Scientific Data 2018 ; Liu et al., arXiv:2102.09542
4	SQuAD v2 (reading comprehension)	DocVQA (~50,000 Q&A pairs)	Dense document OCR and layout reasoning. Directly tests extractive reading comprehension on visual documents. Multimodal document understanding analogue; unlike SQuAD, it adds OCR and layout reasoning.	Mathew et al., arXiv:2007.00398

Implementation Plan

- **Colab T4:**

- *Model:* Qwen2-VL-2B-Instruct (4-bit QLoRA, frozen ViT).
- *Pilot Run:* 1 task (LLaVA-Instruct-150K subset, N=1,000).
- *Core Hypothesis Test:* Does High- G_i induce higher ASR than Moderate- G_i on FigStep and MM-SafetyBench?
- *Audit:* Log $G_i^{(L)}$ *vs.* $G_i^{(P)}$ distributions.

- **PSU Campus Cluster:**

- Run full 4-task continual pipeline across 3 random seeds.
- Evaluate full benchmark suite (MM-SafetyBench, FigStep, JailBreakV-28K, MMHal-Bench).
- Execute ablation over *language LoRA alone* $G_i^{(L)}$, *projector alone* $G_i^{(P)}$, *and joint attribution* $G_i^{(J)}$

Implementation Plan

- **Model Selection:** Qwen2-VL-2B-Instruct shares the Qwen family architecture analyzed in the paper and keeps memory requirements low.
- **LoRA Strategy:** Target q_proj, v_proj, k_proj, o_proj, and gate_proj, up_proj, down_proj in the LLM backbone. Pilot configuration is subject to hardware and implementation validation.
- **Phase 2 Proof-of-Concept Target:**
 - Base Model: Qwen2-VL-2B-Instruct.
 - Dataset: Single task (e.g., 2,000 samples of LLaVA-Instruct-150K or MathVista).
 - Metric to record: ASR on a subset of MM-SafetyBench + Task Accuracy before vs. after fine-tuning using:
 - Standard Fine-Tuning (All samples)
 - Random Selection (20% samples)
 - Moderate- G_i Selection (Ours, 20% closest to median G_i)
 - *Success Criterion:* Confirming that Moderate- G_i preserves lower ASR than Standard/Random fine-tuning while keeping comparable task performance on the VLM.

Safety judges for VLMs

- A text-only judge can evaluate the generated answer, but it may not understand:
 - what harmful instruction was embedded in the image
 - whether the model followed or ignored the visual instruction
 - whether the model misread the image;
 - whether the answer is harmful only in relation to the image.
- Potentially using Llama-Guard-3-8B on the VLM's generated **text responses** for measuring safety output. Need a visual judge to judge the image or the reference harmful instruction, the text prompt, and the the VLM response.
- For visual jailbreak attacks (FigStep, MM-SafetyBench), the judge model evaluates the *final generated text output* against the intended *harmful query embedded in the prompt/image*, not just the raw text prompt.
- For a FigStep-type example, you need to preserve:
 - the original image
 - the intended harmful instruction
 - the VLM response
 - the judge decision
 - whether OCR correctly recovered the instruction

Safety judges for VLMs

- The automatic evaluator could receive a structured record:
 - attack_category
 - image
 - decoded_or_reference_harmful_instruction
 - text_prompt
 - vlm_response
 - judge_label
- Then manually audit a fixed sample.
- A fixed stratified manual audit will verify automated labels across attack categories.

Safety judges for VLMs

- Use a two-stage evaluation:

Stage 1: Automatic judge

- response safety classification
- refusal/compliance label

Stage 2: Manual audit

- attack recognized
- harmful request followed
- visual content understood
- partial compliance
- hallucination

- Store the raw response for every attack, not only the judge score.
- Also report refusal quality. A model that refuses every visual question may have low ASR but poor usefulness. Pair safety with capability:

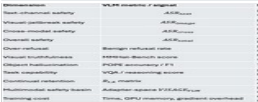
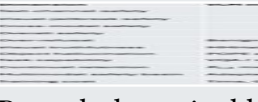
Safety utility tradeoff = (*Low ASR, High TaskScore*)

Threat Model

- The attacker may provide:
 - a harmful text prompt;
 - a harmful instruction embedded in an image;
 - a benign text prompt paired with a malicious image;
 - a benign image paired with a harmful text prompt;
 - an image containing typography, screenshots, diagrams, or rendered text;
 - a jailbreak-style multimodal prompt.
- The attacker does **not** modify model weights during evaluation unless you explicitly study adversarial fine-tuning.
- Then organize attacks by attack surface. Attack evaluation must isolate visual vs. textual channels to detect modality-specific alignment collapse.

Attack condition	Image	Text	Purpose
Text-only harmful prompt	No	Harmful	Decoder safety
Harmful instruction in image	Harmful	Benign	OCR/visual instruction pathway
Harmful image, benign text	Harmful	Benign	Visual semantic safety
Benign image, harmful text	Benign	Harmful	Text-channel safety with visual context
Joint multimodal attack	Harmful or ambiguous	Harmful or ambiguous	Cross-modal interaction

VLM Evaluation and Observability

Dimension	VLM metric / signal	How we will measure it
Text-channel safety		Harmful text prompts with no image; evaluate the VLM’s refusal or harmful completion using an automated judge plus manual audit.
Visual-jailbreak safety		Harmful instructions conveyed through images, including FigStep-style rendered text and visual semantic attacks.
Cross-modal safety		Evaluate paired image-text attacks where the image and text jointly create the harmful instruction.
Overall safety		Aggregate only after reporting the separate text, visual, and cross-modal ASRs. Lower is better.
Over-refusal	Benign refusal rate	Run benign visual questions and safe image-text prompts. Measure how often the model refuses when it should answer.
Visual truthfulness	MMHal-Bench score	Evaluate open-ended visual descriptions and hallucination using the image, question, and generated answer.
Object hallucination	POPE accuracy / F1	Test whether the model incorrectly claims that objects are present in an image.
Task capability	VQA / reasoning score	Evaluate held-out LLaVA-style instruction following, MathVista, medical VQA, and DocVQA test sets.
Continual retention		 
Multimodal safety basin		Perturb the trainable parameter subspace—LM-LoRA, projector, or both, and reevaluate multimodal safety.
Training cost	Time, GPU memory, gradient overhead	Compare standard fine-tuning with sample selection and report wall-clock time, memory, and additional backward passes.

training only adaptors: $\theta^{train} = (\theta_{LM_{LoRA}}, \theta_{projector})$

VLM Continual-Learning Diagnostics

VLM Continual Learning Metrics

Continual-learning metrics

- Let $R_{t,k}$ denote performance on task k after training through task t .
- Display the performance matrix:

	D_1	D_2	D_3	D_4
θ_1	$R_{1,1}$	—	—	—
θ_2	$R_{2,1}$	$R_{2,2}$	—	—
θ_3	$R_{3,1}$	$R_{3,2}$	$R_{3,3}$	—
θ_4	$R_{4,1}$	$R_{4,2}$	$R_{4,3}$	$R_{4,4}$

- Then report: **Backward Transfer** -> Higher is better. Negative BWT indicates degradation on earlier tasks.

$$BWT = \frac{1}{T-1} \sum_{k=1}^{T-1} (R_{T,k} - R_{k,k})$$

- **Forgetting on task k** -> Lower is better.

$$FM_k = \max_{t \in \{k, \dots, T-1\}} (R_{t,k} - R_{T,k})$$

- **Task interference** -> This measures how training on task j changes performance on an earlier task k .

$$I_{j \rightarrow k} = R_{j,k} - R_{j-1,k}$$

VLM specific mechanism Diagnostics

Diagnostic	What to log	Why it matters
Language gradient	$G_i^{(L)}$	Measures response-learning tension in LM-LoRA parameters.
Projector gradient	$G_i^{(P)}$	Measures image-to-language alignment tension.
Joint gradient	$G_i^{(J)}$	Combines language and projector signals after normalization.
Response loss	Mean loss over assistant tokens only	Prevents prompt and image-token positions from contaminating the score.
Target length	Number of assistant-response tokens	Tests whether gradient magnitude is confounded by answer length.
Image complexity	Resolution, aspect ratio, image-token count, OCR density	Tests whether high gradients reflect visual difficulty rather than safety tension.
Parameter drift	Distance from initial adapter/projector weights	Links sample selection to actual model changes.
Safety drift trajectory	ASR and VISAGE at $\theta_0, \theta_1, \dots, \theta_T$	Tests whether safety degradation accumulates across tasks.
Judge validity	Automatic label versus manual audit	Detects errors caused by text-only judging of visual attacks.

VLM Control Conditions and Baselines

Condition	VLM implementation	Scientific question
Aligned checkpoint θ_0	No downstream fine-tuning	What is the initial safety-capability trade-off?
Standard fine-tuning	Train on all available multimodal samples	How much safety drift occurs without selection?
Random 20%	Same training budget and sample count as selection methods	Does any improvement come merely from using fewer samples?
Low- G_i	Select the lowest-gradient multimodal samples	What is the safety-first end of the spectrum?
High- G_i	Select the highest-gradient multimodal samples	Are high-gradient samples specifically associated with safety degradation?
Moderate- G_i	Select samples closest to the median gradient	Does the proposed method preserve safety and capability?
Gradient clipping	Same multimodal training data with fixed clipping threshold	Is controlling update magnitude sufficient?
KL anchoring	Penalize divergence from the aligned VLM on a fixed prompt/image set	Does behavioral anchoring prevent multimodal drift?
EWC or continual-learning regularization	Apply parameter-importance regularization	Is generic forgetting prevention enough to preserve safety?
LM-LoRA-only selection	Rank samples using $G_i^{\{L\}}$	Is the language pathway the main source of risk?
Projector-only selection	Rank samples using $G_i^{\{P\}}$	Is the visual-language bridge the main source of risk?
Joint selection	Rank using normalized $G_i^{\{J\}}$	Does combining both modalities work best?
SafeVLM-style method	Reproduce only if implementation and resources permit	How does data-centric selection compare with architectural safety alignment?

VLM Continual-Learning Diagnostics

Dimension	Metric / signal	How we will measure it
Multimodal safety	$\mathrm{ASR}_{\{\text{text}\}}$, $\mathrm{ASR}_{\{\text{visual}\}}$, $\mathrm{ASR}_{\{\text{cross}\}}$	Evaluate text-only, image-based, and joint image-text attacks separately.
Refusal quality	Harmful refusal rate and benign over-refusal rate	Measure both safe refusal and inappropriate refusal.
VLM safety robustness	$\mathrm{VISAGE}_{\{\text{VLM}\}}$	Perturb the LM-LoRA/projector trainable subspace and reevaluate multimodal safety.
Visual truthfulness	MMHal-Bench and POPE	Measure open-ended hallucination and object-existence errors.
Task capability	MathVista, medical VQA, DocVQA, visual instruction following	Use disjoint held-out test sets after every continual stage.
Task retention	$R_{\{t,k\}}$ performance matrix	Evaluate every previous task after each new task.
Forgetting	BWT and FM	Compute from the task-performance matrix.
Cross-task interference	$I_{j \rightarrow k}$	Measure changes in earlier-task performance after learning task j .
Language gradient	$G_i^{\{L\}}$	Response-only loss gradient over LM-LoRA parameters.
Projector gradient	$G_i^{\{P\}}$	Response-only loss gradient over projector parameters.
Joint gradient	$G_i^{\{J\}}$	Combine normalized language and projector norms.
Sample confounds	Length and image statistics	Log answer length, prompt length, image-token count, resolution, and OCR complexity.
Safety mechanism	Parameter/distribution drift	Compare the adapted VLM with the aligned checkpoint and, if available, the corresponding base model.
Evaluation validity	Judge/manual agreement	Audit automatic safety labels on stratified multimodal examples.
Efficiency	Training time and memory	Measure the extra cost of per-sample gradient computation.

Summary and Open Research questions

- **What We Know (LLMs):** In the source LLM study, high-gradient samples were associated with format mismatch and movement consistent with elastic reversion; median filtering preserves alignment without extra safety data.
- **What We Are Testing (VLMs):**
 - Does multimodal gradient selection preserve VLM safety across both text and visual jailbreaks?
 - Does tension concentrate in the language LoRA weights $[G_i^{(L)}]$ or the cross-modal projector $[G_i^{(P)}]$?
 - Can we eliminate catastrophic forgetting in continual VLM domain adaptation with zero architecture overhead?
 - Can data-centric median filtering achieve competitive safety retention against architectural approaches like SafeVLM without inference penalties?
 - Does high-gradient sample selection drive safety basin collapse in multimodal models as it does in text LLMs?
 - Does any observed improvement remain after controlling for target length, image-token count, and over-refusal?

References

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